# 2021년 한반도 오존존데 일 관측을 통한 화학 재분석자료 검증

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## 1. Introduction

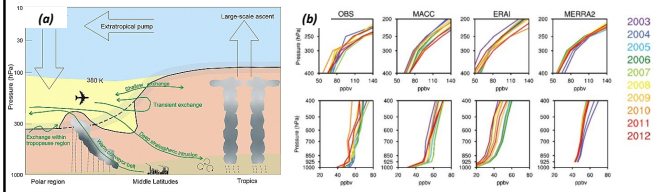



**Figure 1.** (a) Global aspects of STE from Holton et al. 1995, with our new STE concept superimposed (taken from Stohl et al. 2003; their Figure 3), (b) Vertical profile of annual-mean ozone mixing ratio in observation, MACC, ERAI and MERRA2 (taken from Park et al. 2019; their Figure 4).

- **여름철 기간에 한반도 지역의 연직 오존 농도의 변화와 화학 재분석자료 비교를 위해, 전기화학농도셀(Electrochemical Concentration Cell, ECC; Komhyr, 1969) 형식의 오존존데를 사용하여 일단위 관측을 수행**

## 2. Data & Observation

**Table 1.** Date used in this study.

| Type | Data | Variable | Period | Resolution |
|---|---|---|---|---|
| Observation | ECC Ozonesonde | Ozone Partial Pressure, Ozone Mixing Ratio, Total Column Ozone | 2021.08.05 ~ 2021.08.31 | ~ 100 m |
| Atmospheric reanalysis | ERA5 | Geopotential, Potential Vorticity, U Velocity, V Velocity | | 1.5° × 1.5° |
| Chemical reanalysis | MERRA-2 | Ozone Mixing Ratio, Total Column Ozone, Pressure, U Velocity, V Velocity, Temperature | | 0.5° × 0.625° |
| | CAMS | | | 0.75° × 0.75° |

| Data | Publish | Ozone Source | Reference |
|---|---|---|---|
| MERRA-2 | 1 Oct 2004 - present | MSL, OMI, SBUV and SBUV/2 | Gelaro et al., 2017 |
| CAMS | 2003 - present | SCIAMACHY, MIPAS, MLS, OMI, GOME-2 and SBUV/2 | Inness et al., 2019 |

ERA5: ECMWF Reanalysis v5
MERRA-2: Modern-Era Retrospective analysis for Research and Applications, Version 2
CAMS: Copernicus Atmosphere Monitoring Service

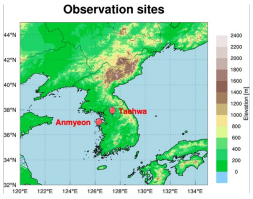



**Figure 3.** Locations of Anmyeon (lat: 36.54°N, lon: 126.33°E) and Teahwa (lat: 37.31°N, lon: 127.31°E) ozonesonde observation sites in South Korea. Shading denotes elevation(unit: m).

- 총 26개 고해상도 오존 프로파일 생성 (안면도 24개, 태화산 2개)
- 화학 재분석자료는 관측지점과 가장 가까운 격자를 사용하여 비교 분석

## 3-1. Observation Result

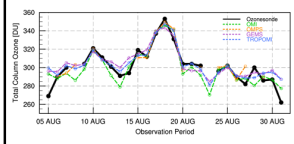



**Figure 4.** Time series of total column ozone (TCO; unit: $DU$) from ozonesonde measurements (black) and satellites (OMI: green, OMPS: orange, GEMS: purple, and TROPOMI: blue)

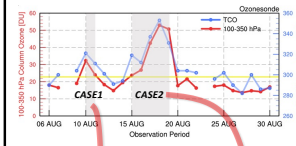



**Figure 5.** Time series of ozonesonde total (red) and upper tropospheric (blue, 100-350 $hPa$) column ozone (CO, unit: $DU$). Gray shading indicates stratosphere-troposphere exchange (STE) periods defined by criterion of CO > average or 100-350 $hPa$ CO (22.8 $DU$, yellow line).

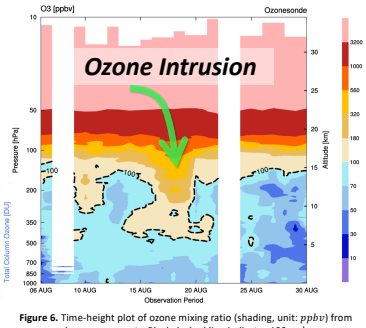



**Figure 6.** Time-height plot of ozone mixing ratio (shading, unit: $ppbv$) from ozonesonde measurements. Black dashed line indicates 100 $ppbv$.

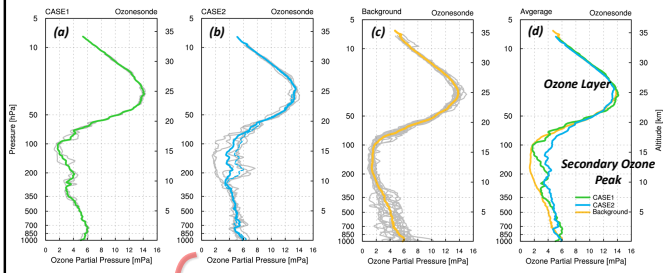



**Figure 7.** Profiles of ozone partial pressure (thin gray lines, unit: $mPa$) from ozonesonde measurements for (a) CASE1, (b) CASE2, (c) BACKGROUND, and (d) all cases. The mean profile of each set is shown in a thick line(green: CASE1, blue: CASE2, orange: BACKGROUND).

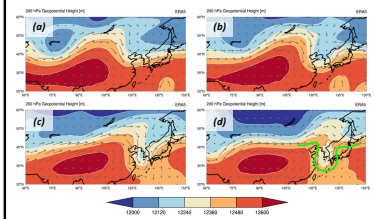



**Figure 8.** 200-hPa geopotential height (shading, unit: $m$) and wind (gray vector, unit: $m\ s^{-1}$) from ERA5 reanalysis at 06 UTC (a) 16, (b) 17, (c) 18, and (d) 19 Aug 2021. The green line in (d) denotes anticyclonic wave breaking (AWB).

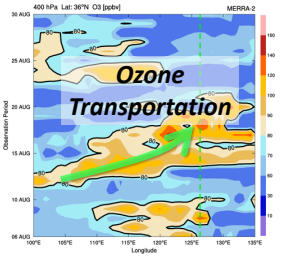



**Figure 9.** The Hovmöller diagram of the ozone mixing ratio (shading, unit: $ppbv$) from MERRA-2 at 400 hPa. Black solid line indicates 80 $ppbv$, green dashed line indicates Anmyeon station longitude (126.33°E).

## 3-2. Intercomparison Reanalysis Data

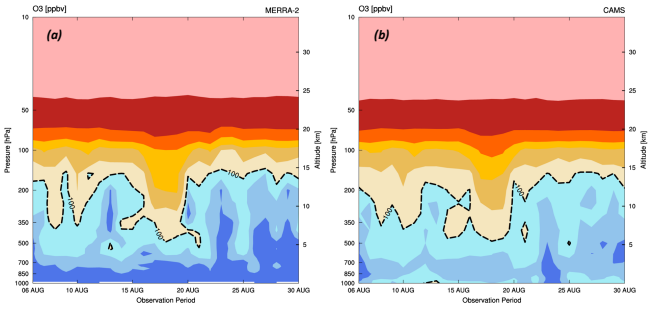



**Figure 10.** Same as in Figure. 6, but from chemical reanalyses of (a) MERRA-2 and (b) CAMS.

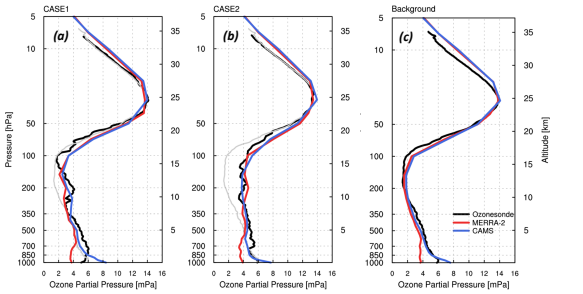



**Figure 11.** Mean profiles of ozone partial pressure (unit: mPa) from ozonesonde measurements (black), MERRA-2 (red), and CAMS (blue) for (a) CASE1, (b) CASE2, and (c) BACKGROUND. The gray lines in (a) and (b) denote mean profiles of BACKGROUND from ozonesonde measurements.

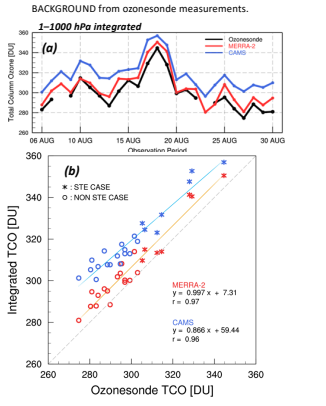



**Figure 11.** (a) Time series of total column ozone from ozonesonde measurements (black), MERRA-2 (red), and CAMS (blue). (b) Scatter plot with regression lines of total column ozone from MERRA-2 (red) and CAMS (blue) against ozonesonde measurements. Asterisk markers represent the stratosphere-troposphere exchange case. The gray dashed line represents the 1:1 ratio line. The least squares regression equations and the correlation coefficients (r) are also presented.

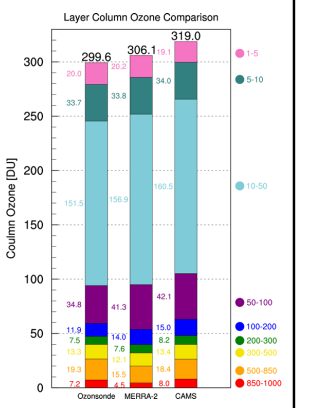



**Figure 12.** Bar plot of accumulated mean column ozone from ozonesonde measurements (left), MERRA-2 (middle), and CAMS (right). For each color box denotes layers used to integrate column ozone. Number on the left from the bar presents the value of column ozone from layers. Number on top of the bar presents the total column ozone.

## 4. Summary

- 2021년 8월 5일부터 31일까지, 서해안에 위치한 안면도를 중심으로 오존존데 일단위 관측을 수행하여, 총 26개 고해상도 오존 프로파일을 생성.
- STE 발생 당시 오존 연직 구조가 Secondary Ozone Peak 연구(Hwang et al., 2007)에서 보이는 연직 구조와 매우 유사하게 나타남.
- TCO의 상승에는 성층권에서부터 오존 유입뿐만 아니라, 대류권에서 수송에 의한 증가도 있음.
- 화학 재분석자료에서 STE 구조를 잘 나타내지만, 정량적으로는 차이가 보임.
- 1-1000 hPa 까지의 TCO을 계산시, 일 변동성은 매우 유사하나 두 화학 재분석자료 모두 과대모의함.
- 대류권 하층에서는 MERAA-2는 과소모의, CAMS는 오존존데 관측값과 유사하나, 200 hPa 이상부터는 두 재분석자료 모두 오존존데에 비해 과대모의함.

## 5. Reference

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